VRIJE UNIVERSITEIT BRUSSEL

ENVIRONMENTAL PROGRAMMING

ASSIGNMENT 2

Carbon Bombs, Analyzing the Potential Exposure to Climatic Extremes Induced by the Largest Fossil Fuel Projects

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1 Introduction

Carbon bombs are the 425 largest fossil fuel extraction mega-projects, each with a potential emission of at least 1 $GtCO_2$ [1]. A recent update of this database, accounting for the reserves extracted since 2020, yields 912 (878–948) $GtCO_2$, more than seven times the remaining 1.5°C carbon budget of 130 $GtCO_2$ [2]. If fully released, these CO_2 emissions alone would add about 0.4°C of global warming, while the fugitive methane released during extraction (2.52 $GtCH_4$) would add a further 0.08–0.24°C of peak warming [2]. The largest carbon bomb, the *Permian Delaware Tight* shale oil and gas field in Texas, USA, still contains about 23 $GtCO_2$ in the 2025 update of the database [2], equivalent to about 60% of global fossil CO_2 emissions in 2025 [3]. Our climatic future therefore crucially depends on the decisions made regarding the exploitation of these projects.

Given their massive potential greenhouse gas emissions, these carbon bombs are expected to substantially affect the frequency and intensity of climate extremes, such as droughts, heatwaves and floods, experienced by current and future generations [2, 4, 5]. Among the 425 projects identified, about 60% are already in operation, while 170 have been planned but not yet developed. This highlights the urgent need to assess the climate impacts that would result from the continued development or the cancellation of these projects. A recently developed framework makes it possible to compute, for an individual source of greenhouse gas emissions such as the CO_2 emissions of a carbon bomb, the number of people who will be exposed to an additional climate extreme over their lifetime as a result of these emissions [2].

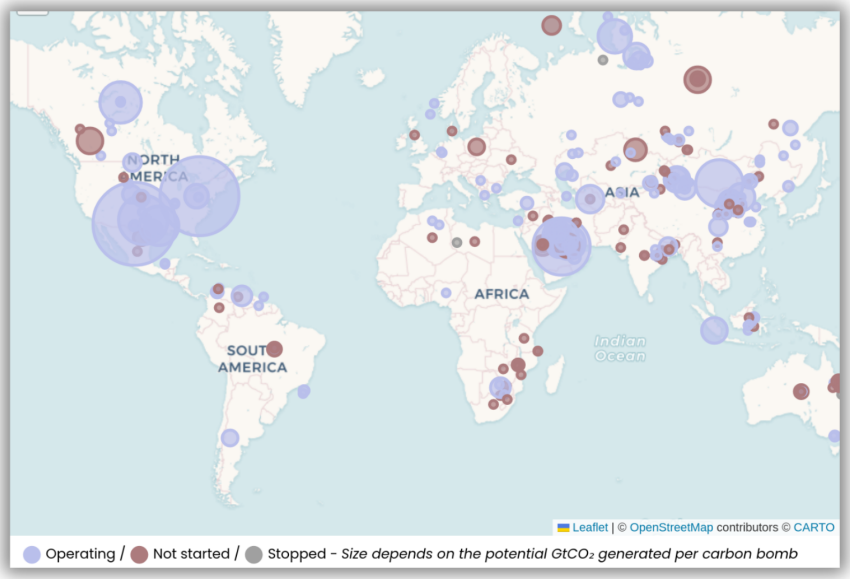


Figure 1: Map of carbon bombs worldwide. Figure from `https://carbonbombs.org/`

The purpose of this project is to analyse the updated carbon bombs dataset, together with datasets containing the number of people born in 1950, 1990, 2010 and 2025 in 183 countries who are expected to experience at least one additional climate extreme due to the CO_2 emissions of the *Permian Delaware Tight* carbon bomb. The climate extremes considered in this analysis are heatwaves, droughts, crop failures, river floods, wildfires and tropical cyclones.

For this purpose, you will learn or improve your skills in:

- Handling and analysing typical spreadsheet .
- Handling and analysing large climate impact datasets.
- Using a `Jupyter Notebook` to analyse and visualise data.
- Working with `DataFrame` and `Dataset` structures.
- Using libraries such as `numpy`, `pandas`, `geopandas` and `xarray`.
- Creating maps using `GeoDataFrame` objects and `matplotlib`.

2 Assignment

2.1 Datasets

All the data needed for this project can be downloaded from the project directory.

- `carbon_bombs_co2_dataset_kuhne_2022`: original carbon bomb dataset from Kühne et al. [1].
- `carbon_bombs_co2_dataset_zhang_2025`: updated carbon bomb dataset from Laridon et al. [2].
- `ds_Carbon_Bombs_CO2_394_Permian_Delaware_Tight_all_hazards_all_regions_XXXX`: exposure data for the XXXX birth cohort in all regions, due to the CO_2 emissions of the Permian Delaware Tight carbon bomb.
- `ds_Carbon_Bombs_CO2_394_Permian_Delaware_Tight_all_hazards_all_countries_XXXX`: exposure data for the XXXX birth cohort in all countries, due to the CO_2 emissions of the Permian Delaware Tight carbon bomb.
- `gdf_country_borders.pkl`: `GeoDataFrame` containing the geometric objects associated with country borders.

In the file names above, `XXXX` stands for the birth year of the cohort (1950, 1990, 2010 or 2025).

2.2 Tasks

1. Create a Git repository and an associated GitHub repository for your project.
   (a) Add a `README.md` file, which you will complete at the end of the project with all the information a future user needs to run your scripts and use your data.
   (b) Create sub-folders named `data/`, `notebooks/` and `figures/` to store the corresponding files.
   (c) Export the Python environment needed to run your scripts with `conda` to a file named `environment.yml`.
2. Download the updated carbon bomb dataset (`carbon_bombs_co2_dataset_-zhang_2025`), which is a `.xlsx` file. Open the `coal_combustion` and `oil&gas_combustion` sheets and make sure you understand the information they contain.
3. Create a Jupyter notebook named `carbon_bombs_analysis.ipynb`. In the first markdown cell, write a comprehensive description of the purpose of the notebook and the associated references.
4. Create a function `pre_processing_carbon_bombs()` that will:
   (a) Open your `.xlsx` file, read the two sheets mentioned above and combine them into a single dataframe.
   (b) Retain only the columns useful for the project: `"Project"`, `"Country"`, `"Potential emissions (Gt CO2)"` (average, low and high estimates) and `"Fuel"`.
   (c) Convert the values of `"Potential emissions (Gt CO2)"` to floats.
   (d) Open the original dataset from Kühne et al. [1] and use its `Full Carbon Bombs List` sheet to assign to each project of your dataset its project ID, as well as a new variable called `new`, equal to 1 if the project was marked with a `*` in the original dataset and 0 otherwise.
   (e) Set the project `name` as the index of your dataframe.
   (f) Construct a dataset `ds_carbon_bombs` with three coordinates: `name` for the project name, `country` for the country of origin and `fuel` for the type of fossil fuel associated with the carbon bomb. Then, add six variables to this dataset: `potential_emissions` (with its low and high estimates), `new`, `total`, `total_oil_and_gas`, `total_coal` and `total_new_carbon_bombs`. It is useful to define attributes for your dataset to explain, using metadata, what it contains and the units of the stored values. Adding one attribute per variable is a plus for clarity. Finally, save your dataset in `pickle` format in your `data/` folder.

5. Create a bar plot showing the potential emissions in $GtCO_2$ of the 20 largest carbon bomb projects, based on your `ds_carbon_bombs` dataset. Make sure that the name of each carbon bomb project is displayed on the x-axis. To provide more information, use the `country` coordinate of your dataset to give each country a different colour in the bar plot. You can also use the `fuel` coordinate to hatch the bars or not, in order to distinguish the type of fossil fuel among these 20 carbon bombs. Use the low and high estimates to add error bars.

6. Your `ds_carbon_bombs` dataset is very rich and allows for many figures that help to better understand the data behind these carbon bombs. There are no additional specific instructions, but if you have time, it may be useful to produce further figures showing, for instance, the spatial distribution of the projects or the cumulative potential emissions by country or by type of fossil fuel. Let your creativity guide you!

7. Download the remaining datasets associated with the Permian Delaware Tight project: `ds_Carbon_Bombs_CO2_394_Permian_Delaware_Tight_all_hazards_all_regions_XXXX` and `ds_Carbon_Bombs_CO2_394_Permian_Delaware_Tight_all_hazards_all_countries_XXXX`. Load these datasets for every available birth cohort into your notebook and print them to understand their structure. For the Permian Delaware Tight carbon bomb, they contain projections of the number of people born in different years, per country and per region, who will experience an additional climate extreme due to its emissions. For each spatial resolution, merge the datasets of the different birth cohorts into a single dataset with a new coordinate named `birth_years`.

8. Create a function to plot a map of the `median_val`, representing the number of children born in 2025 who are additionally exposed, per region. To achieve this, you will need to create an object `gdf_regions_border` of type GeoDataFrame. To construct this object, you can use the file `gdf_country_borders.pkl`, which contains the precomputed `GeoDataFrame` at the country level and is available in the project directory.

9. Create a function to plot a map of the `median_val` of the number of children born in 2025 who are additionally exposed, per country. For this, you will need the object `gdf_country_borders.pkl`, which you can download from the project directory.

10. For the regional map, use the variables `ci_lower` and `ci_upper` to add a box for each region, highlighting the best estimate (`median_val`) and the 90% uncertainty range given by `ci_lower` and `ci_upper`. Apply the same idea to the country-level map, but only for the five most affected countries, and include Belgium for comparison.

11. Save these two figures as high-resolution `.png` files in your `figures/` folder.

12. Use the results at both spatial resolutions for the different birth cohorts to produce outputs and figures that highlight the intergenerational inequity in additional lifetime exposure to climate extremes resulting from the CO_2 emissions of the Permian Delaware Tight carbon bomb.

13. The regional and country-level datasets also allow for the visualisation of much other interesting information. For instance, each dataset contains a variable called `S2S_LH_results`, which holds a statistical sample of the number of people exposed to one additional hazard. The variables `median_val`, `ci_lower` and `ci_upper` are computed from this sample, so many statistical analyses of the results are possible. Consider which aspects of the data could be useful to visualise in additional figures.

14. Once you have completed all these steps in the notebook `carbon_bombs_analysis.ipynb`, you will have created a notebook that allows users to follow, step by step, the process you have implemented. Based on these code cells, create a Python script `carbon_bombs_analysis.py` in which you define functions for the different code blocks, and a final function `carbon_bombs_analysis()` that calls the previously defined functions to execute them.

For the clarity of your work, it is important to keep a well-organised structure in your notebook, with a cell loading all modules at the top, and markdown titles and subtitles for the different steps of your data processing and visualisation. Do not forget to comment your code thoroughly.

2.3 Optional Tasks

1. Explore the datasets further and consider creating other figures and computing other statistics to enhance your understanding and analysis.

2. Develop a simple graphical user interface (GUI) to run your code interactively. Tip: you are free to use any open-access Python package and to design the layout of the interface as you wish. We suggest trying Tkinter (`https://wiki.python.org/moin/TkInter`), but this is just a suggestion; many other options are available.

   Once again, the internet is an excellent source of information (documentation, videos, tutorials, etc.), e.g. `https://www.geeksforgeeks.org/python/create-first-gui-application-using-python-tkinter/`. A few useful websites and YouTube videos:

   - `https://www.youtube.com/watch?v=eJRLftYo9A0&list=PLQVvvaa0QuDclKx-QpC9wntnURXVJqLyk&index=8`

- `http://sebsauvage.net/python/gui/`
- `https://www.tutorialspoint.com/python/python_gui_programming.html`
- `https://docs.python.org/3/library/tkinter.html`
- `http://www.tkdocs.com/tutorial/index.html`

2.4 Libraries

These are the Python libraries you will most likely need to complete the tasks: `xarray`, `pickle`, `matplotlib`, `os`, `cartopy`, `geopandas`, `sys`, `numpy`, `pandas`, `scipy`, `regionmask` and `textwrap`.

3 Submission and Report

Develop software that solves the steps described in the previous section, working collaboratively with GitHub Desktop (or the GitHub CLI). Submit a link to your GitHub repository to Elga Salvadore (elga.salvadore@vub.be) and Amaury Laridon (amaury.laridon@vub.be) by **21st December 2026**. Make sure the online GitHub repository contains: (1) all the Python codes you have developed to complete the tasks described in Section 2.2, (2) a Jupyter Notebook to present and run your code (in the style of a user manual or tutorial), (3) a section of the Jupyter Notebook describing the use of generative AI (including sample of questions), (4) a section of the Jupyter Notebook describing the contributions of each team member (if not clear from the GitHub contributions), (5) a flow chart of your code (which can be included in the Jupyter Notebook), and (6) a `.yml` file of the environment you created for your project.

3.1 Need a Clarification?

Please contact Amaury Laridon (Amaury.Laridon@vub.be) and Elga Salvadore. You can pose your questions via email and request in-person and/or online meetings.

3.2 How is the Assignment Evaluated?

EVALUATION MATRIX Assignment	Point
1) Does your software perform the all the required tasks providing the correct result(s)?	10 points
2) Is your code readable?	± 1 point
3) Are the name of the variables\functions\modules meaningful?	± 1 point
4) Are the data types appropriate for values the hold?	± 0.5 point
5) Are there too many code repetitions?	± 0.5 point
6) Does your code have a coherent structure?	± 0.5 point
7) Is the Jupyter Notebook clear, to the point and complete? Did you create a flow chart?	± 1 point
8) Is your code easy to use?	± 1 point
9) Is there a good balance of comments and code in the scripts?	± 0.5 point
10) Code reuse: Did you forget to properly citing the source of codes you are reusing?	- 2 point
11) How flexible is your code? For example, if I use a different data set with a different file name, would it still be working?	± 1 point
12) Use of generative AI: Were you transparent in the use of generative AI? Are the reported questions sufficiently detailed? Did you test the code that was generated by generative AI?	± 1 point
13) Did you develop a GUI for your code?	Up to 2 points

References

[1] Kjell Kühne, Nils Bartsch, Ryan Driskell Tate, Julia Higson, and André Habet. "Carbon Bombs" - Mapping key fossil fuel projects. *Energy Policy*, 166:112950, July 2022.

[2] Amaury Laridon, Rosa Pietroiusti, Chris Smith, Harry Zekollari, Lilian Schuster, Alexander Nauels, Matthew Palmer, Jacob Schewe, Jiayi Zhang, Carl-Friedrich Schleussner, Inga Menke, Joeri Rogelj, Rupert Stuart-Smith, and Wim Thiery. A framework linking project-level greenhouse gas emissions to additional climate extremes exposure and multi-century committed change. in prep., 2026.

[3] Global Carbon Budget 2025. *Earth System Science Data*, 18(5).

[4] Nerilie J. Abram, Nicola Maher, Sarah Perkins-Kirkpatrick, Georgina M. Falster, Terry P. Hughes, Katrin J. Meissner, Louise J. Slater, Andrew D. King, Andrew J. Pitman, Gillian Moon, and Wesley Morgan. Quantifying the regional to global climate impacts of individual fossil fuel projects to inform decision-making. *npj Climate Action*, 4(1):92, October 2025. Publisher: Nature Publishing Group.

[5] Luke Grant, Inne Vanderkelen, Lukas Gudmundsson, Erich Fischer, Sonia I. Seneviratne, and Wim Thiery. Global emergence of unprecedented lifetime exposure to climate extremes. *Nature*, 641(8062):374–379, May 2025. Publisher: Nature Publishing Group.